The Depleted Safety Net: Provider Attrition and Medicaid's Pandemic-Era Disruption*

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Abstract

Using provider-level Medicaid claims for 617,000 billing entities across all 50 states over seven years, I test whether pre-pandemic provider attrition predicted COVID-era disruption in home and community-based services (HCBS). The answer depends on measurement. A purely pre-treatment exit rate (2018-active providers absent from all 2019 billing) yields a null static difference-in-differences result (0.614, $p = 0.56$). A broken-trend model reveals that high-attrition states were already declining before COVID ($\lambda = -0.029$, $p = 0.025$), experienced no pandemic-onset level shift ($\beta \approx 0$), and showed post-COVID trajectory improvement ($\kappa = 0.033$, $p = 0.024$). By contrast, a post-treatment-contaminated exit rate yields significant effects (-0.844, $p = 0.017$)—a mechanical artifact of embedding the pandemic shock into the treatment variable. Pre-pandemic attrition predicts ongoing decline but not pandemic amplification; measuring "exits" in administrative data requires careful attention to the observation window.

JEL Codes: I11, I13, I18, J44, H75

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1. Introduction

A basic prediction of competitive labor markets is self-correction. Workers exit; wages adjust; new workers enter. The equilibrium may shift, but the market clears. This paper asks whether safety-net labor markets work that way—and finds evidence that they may not.

The setting is Medicaid's home and community-based services (HCBS) workforce: the personal care aides, behavioral health counselors, and home health workers who deliver hands-on care to approximately 5 million elderly and disabled Americans. Between 2018 and early 2020, this workforce was quietly eroding. Providers were exiting the Medicaid billing system at rates that varied dramatically across states—from 10 percent of the active 2018 provider base absent from all 2019 billing to over 55 percent in others. Then COVID-19 arrived—but did it make the erosion worse?

Using the newly released Transformed Medicaid Statistical Information System (T-MSIS) provider spending dataset—the first public provider-level claims data covering HCBS, spanning all 50 states and the District of Columbia from January 2018 through December 2024 (with the primary analysis truncated at June 2024 to avoid reporting lag contamination)—I construct a purely pre-treatment measure of provider exit intensity: the share of 2018-active providers absent from all 2019 billing. This measure, free from post-treatment contamination, yields a null static result: the DiD coefficient is 0.614 (SE = 1.040, $p = 0.56$). A collapsed cross-sectional regression (N = 51 states) confirms the null ($p = 0.62$ under HC2 standard errors). However, the broken-trend model reveals significant structure: an ongoing pre-COVID differential decline in high-attrition states ($\lambda = -0.029$, $p = 0.025$), no level shift at March 2020 ($\beta \approx 0$), and a significant post-COVID slope improvement ($\kappa = 0.033$, $p = 0.024$) suggesting partial mean-reversion. The pandemic did not amplify pre-existing attrition; if anything, the post-COVID trajectory improved relative to the pre-trend.

A critical methodological contribution emerges from comparing exit rate definitions. Prior versions of this paper used a pandemic-contaminated measure—providers active in 2018–2019 but absent after February 2020—which yielded a significant negative coefficient (-0.844, $p = 0.017$). Reviewers correctly identified that this measure embeds post-treatment information: providers who stopped billing because of COVID are classified as "pre-pandemic exits," mechanically producing the observed association. This revision replaces the treatment variable with a purely pre-treatment measure (2018-active, absent in all 2019 billing) and finds that the significance disappears entirely. The broken-trend model clarifies: high-attrition states were already declining before COVID, and the pandemic did not accelerate this decline. Five alternative randomization inference stratifications (unconditional $p = 0.143$; Census divisions $p = 0.278$; Census regions $p = 0.163$; urbanicity $p = 0.179$; governor party $p = 0.185$)

all fail to reject the null. The contribution is methodological: measurement of "provider exits" in administrative claims data is highly sensitive to the observation window, and post-treatment contamination can generate spurious causal findings.

The economic implications cut two ways. The null finding on pandemic amplification challenges the hysteresis narrative: Blanchard and Summers (1986) and Yagan (2019) document how temporary shocks can permanently alter equilibria, but the broken-trend results suggest that the HCBS workforce decline was already underway before COVID and that the pandemic may have moderated—not amplified—the trajectory. The positive κ is consistent with policy response: Congress invested \$37 billion through the American Rescue Plan Act (ARPA) starting in April 2021, and this injection may have partially arrested the pre-existing decline in high-attrition states. Nevertheless, the ongoing pre-trend ($\lambda < 0$) documents persistent erosion of the HCBS safety net that predates the pandemic and warrants policy attention regardless of the causal role of COVID. The contribution of this paper is thus twofold: a substantive finding about ongoing HCBS workforce decline, and a cautionary methodological lesson about post-treatment contamination in administrative exit rate measures.

The null finding is not specific to HCBS. The non-HCBS falsification coefficient (0.512, $p > 0.60$) is similarly insignificant, confirming that the pre-period exit rate does not predict pandemic-era disruption for any Medicaid provider type. The contribution of this paper is therefore methodological as much as substantive: the HCBS workforce was declining before COVID, and measurement choices—specifically, whether exit rates embed post-treatment information—determine whether this decline appears causally linked to the pandemic.

This paper contributes to three literatures. The literature on healthcare supply and access has established that provider availability shapes utilization and outcomes (Finkelstein et al., 2012; Baicker et al., 2013), but has focused overwhelmingly on the demand side—insurance coverage, cost-sharing, eligibility rules. I document that the supply side matters independently: beneficiaries who are already covered and enrolled lose access when no provider exists to deliver care. On the workforce side, the COVID-era literature has documented acute staffing disruptions in hospitals (Sinsky et al., 2021) and nursing homes (Alexander and Schnell, 2021), but has not connected pre-existing workforce conditions to pandemic-era outcomes. I use the T-MSIS data to trace the chain from supply (provider counts) through access (beneficiary-provider encounters) to intensity (claims per encounter), documenting that the weakest link in the safety net was forged years before the pandemic. The safety-net resilience literature, from Dranove et al. (2003) on hospital vulnerability to Duggan (2000) on Medicaid payment structure, has emphasized how institutional features create fragility. I extend this logic to the labor market that underlies service delivery: when the workforce erodes gradually through attrition, the system becomes brittle—and a shock of sufficient

magnitude exposes that fragility.

Section 2 provides institutional background. Section 3 develops the conceptual framework. Sections 4 and 5 describe the data and empirical strategy. Section 6 presents results, and Section 7 discusses implications.

2. Institutional Background

2.1 Home and Community-Based Services in Medicaid

Medicaid is the largest single payer for long-term services and supports (LTSS) in the United States, spending approximately $190 billion annually on services for elderly and disabled beneficiaries (Centers for Medicare and Medicaid Services, 2024). A growing share of this spending flows through home and community-based services (HCBS)—personal care, habilitation, behavioral health, and attendant care delivered in beneficiaries' homes rather than in institutional settings. This shift toward community-based care, driven by the 1999 *Olmstead v. L.C.* Supreme Court decision and subsequent federal policy, has made the HCBS workforce the linchpin of the Medicaid safety net.

The *Olmstead* decision required states to serve individuals with disabilities in the most integrated setting appropriate to their needs. In the quarter-century since, HCBS spending has grown from roughly 20 percent of Medicaid LTSS expenditures to over 60 percent in most states. By 2019, approximately 5 million Medicaid beneficiaries received HCBS—a number that rivals the combined enrollment of all nursing homes in the country. This rebalancing has generated enormous demand for a workforce that the labor market has struggled to supply.

The HCBS workforce differs fundamentally from the medical workforce studied in most health economics research. Personal care aides and home health workers earn a median hourly wage of $14.15 (Bureau of Labor Statistics, 2023), face annual turnover rates of 40–60 percent (PHI, 2023), and rarely receive employer-sponsored health insurance. Many are independent contractors or employees of small agencies rather than large healthcare systems. These structural features make the HCBS workforce exceptionally fragile: low wages and poor working conditions produce chronic shortages even in normal times, and small providers can disappear entirely when a single worker retires or relocates.

The fragility is compounded by the structure of Medicaid reimbursement. Unlike Medicare, which sets national fee schedules, Medicaid rates are determined state by state. As of 2019, the median state Medicaid reimbursement rate for personal care services was approximately $18 per hour—well below the rate needed to attract and retain workers when Amazon warehouses and fast-food restaurants offered $15–$17 per hour with benefits. States with the lowest reimbursement rates tended to have the highest provider turnover, creating a vicious cycle:

underpayment drives exits, exits reduce access, and reduced access discourages new entrants who see limited career prospects in a shrinking market.

State-level variation in HCBS program design further shapes the provider landscape. Some states operate HCBS primarily through 1915(c) waivers, which cap enrollment and create waiting lists; others use 1915(k) Community First Choice or 1115 demonstration waivers that provide broader access. States also differ in whether they allow self-directed services (where beneficiaries hire their own workers) versus requiring agency-based provision. These structural differences affect both the size and composition of the provider workforce: self-directed programs tend to have more individual providers and higher turnover, while agency-based programs have fewer but larger organizational providers.

HCBS spending is identifiable in Medicaid claims through distinctive procedure codes. The Healthcare Common Procedure Coding System (HCPCS) assigns T-prefix codes to HCBS services (e.g., T1019 for personal care per 15 minutes), H-prefix codes to behavioral health services (e.g., H2016 for community support per diem), and S-prefix codes to temporary Medicaid services (e.g., S5125 for attendant care per 15 minutes). These codes have no Medicare equivalent—they exist solely within the Medicaid system. As the T-MSIS data reveal, T/H/S codes account for approximately 52 percent of total Medicaid provider spending, making them the dominant category in provider-level claims.

2.2 The HCBS Workforce Crisis Before COVID-19

The pandemic did not create the HCBS workforce crisis—it exposed and deepened one that had been building for years. Prior to 2020, multiple factors were eroding the provider base. Low Medicaid reimbursement rates, which had been frozen or declining in real terms in many states, made HCBS work economically unattractive relative to retail and fast-food jobs that offered comparable pay with less physically and emotionally demanding work (PHI, 2020). An aging workforce meant that experienced providers were retiring at accelerating rates. And consolidation in home health—driven by managed care contracting and regulatory requirements—was eliminating small providers who could not bear administrative overhead.

The demographic composition of the exiting workforce deserves attention. HCBS providers skew older and more female than the general healthcare workforce: the median home health aide is in her mid-40s, and over 85 percent of direct care workers are women. Providers enumerated in the earliest years of the NPI system (2006–2008, when the mandate took effect) had, by 2019, accumulated 11–13 years of practice—long enough for burnout and physical wear to drive retirement decisions. The NPPES data confirm that providers with earlier enumeration dates exit at substantially higher rates, consistent with career-cycle dynamics rather than market-wide shocks.

Geographic concentration of exits was also pronounced. States in the Southeast and Mountain West, which tend to have lower Medicaid reimbursement rates and fewer large healthcare systems, experienced exit rates 50–100 percent higher than states in the Northeast and Pacific Coast. This pattern suggests that exit was not random but systematically related to the adequacy of state Medicaid investment—a relationship that becomes central to the identification strategy.

These trends are visible in the T-MSIS data. Of the approximately 617,000 billing entities that appear in the raw dataset (January 2018 through December 2024, 84 months), only 6 percent bill continuously across all months. Over 38 percent appear for fewer than 12 months. The median billing tenure is 22 months. This extraordinary dynamism—far greater than turnover in physician or hospital markets—reflects the structural fragility of the HCBS sector.

2.3 COVID-19 and the HCBS Sector

The pandemic struck the HCBS sector through multiple channels simultaneously. Lockdowns and social distancing orders disrupted in-person service delivery. Infection risk deterred workers from entering beneficiaries' homes and vice versa. Emergency Medicaid expansions—driven by the Families First Coronavirus Response Act's continuous enrollment requirement—increased the number of beneficiaries seeking services, while the supply of available workers contracted. Many HCBS workers, lacking personal protective equipment and paid sick leave, were forced to choose between exposure and income.

The interaction between demand surge and supply contraction was particularly acute. Medicaid enrollment grew by approximately 20 million between February 2020 and its peak in early 2023, driven by both economic displacement and the continuous enrollment requirement. Yet the HCBS workforce was shrinking, not growing. The resulting gap—more beneficiaries, fewer providers—translated directly into longer wait times, reduced service hours, and in some cases complete loss of access to home-based care. For beneficiaries with severe disabilities or chronic conditions, the absence of HCBS providers meant heightened risk of hospitalization, institutionalization, or informal caregiving arrangements that imposed substantial burdens on family members.

The effects were severe and uneven across states. States with stricter lockdowns experienced larger initial disruptions to HCBS delivery (Kaiser Family Foundation, 2022). States with lower pre-pandemic Medicaid reimbursement rates saw more provider exits, as workers left for better-compensated essential worker positions in retail and warehousing. Competition from the booming gig economy—ride-sharing, food delivery, and online retail—further accelerated attrition from the HCBS sector, which could not match the flexibility and often

comparable wages offered by these alternatives. And states that entered the pandemic with already-depleted provider networks—the focus of this paper—had less slack to absorb these shocks.

2.4 The American Rescue Plan's HCBS Investment

In response to the cascading HCBS crisis, Congress enacted the American Rescue Plan Act (ARPA) in March 2021. Section 9817 provided a temporary 10 percentage point increase in the Federal Medical Assistance Percentage (FMAP) for HCBS expenditures, effective from April 1, 2021, through March 31, 2022. States were required to use the federal savings to "supplement, not supplant" existing HCBS spending—channeling funds toward rate increases, workforce recruitment and retention bonuses, service expansions, and infrastructure improvements.

All 50 states and the District of Columbia participated in the enhanced FMAP. The total federal investment amounted to approximately $37 billion, with $26 billion directed toward workforce-related activities (ADvancing States, 2024). However, the timing and nature of state spending varied substantially. Some states implemented provider rate increases within months of CMS approval; others took over a year to deploy funds. Some states focused on permanent rate increases; others relied on one-time bonus payments. This variation in implementation, combined with the uniform April 2021 start date for the enhanced FMAP, provides the empirical setting for evaluating ARPA's effectiveness.

The enhanced FMAP expired on March 31, 2022, but states had until March 2024 (later extended to March 2025 for 13 states) to spend accumulated savings. This extended spending window means that ARPA-funded interventions were being implemented throughout the 2021–2024 period, making the post-April 2021 treatment window appropriate for identifying supply-side effects.

3. Conceptual Framework

Consider a state s with a Medicaid HCBS provider market. Let N_{st} denote the number of active providers at time t, governed by entry and exit flows:

$$N_{st} = N_{s,t-1} + \text{Entry}_{st} - \text{Exit}_{st} \tag{1}$$

In steady state, entry roughly equals exit. But if exit accelerates—due to retirements, wage competition, or burnout—and entry does not adjust, the stock N_{st} declines. Define the *pre-period exit rate* θ_s as the share of providers active in 2018 who are absent from all 2019

billing:

$$\theta_s = \frac{\#\{\text{providers with} \geq 1 \text{ claim in 2018, zero claims in all of 2019}\}}{\#\{\text{providers with} \geq 1 \text{ claim in 2018}\}} \tag{2}$$

This measure captures the degree to which the provider network was already eroding before the pandemic arrived. It is *pre-determined* with respect to the pandemic shock—constructed entirely from 2018–2019 billing data, with no reference to post-COVID outcomes. No provider exiting in 2018 or 2019 did so in anticipation of COVID-19.

The pandemic creates a common adverse shock to all provider markets, but its impact is amplified in markets with higher pre-existing depletion. Formally, let ΔY_{st} be the change in a service outcome after March 2020. The prediction is:

$$\frac{\partial \Delta Y_{st}}{\partial \theta_s} < 0 \tag{3}$$

States with higher θ_s experience larger declines because they have less network "slack"—fewer providers to absorb increased demand, cover for absent workers, or maintain service continuity during disruptions.

The prediction applies to multiple outcomes along the supply-access chain. At the supply level, higher θ_s reduces N_{st} (active providers). But the consequences propagate downstream: with fewer providers, beneficiaries face longer wait times and reduced service availability, leading to declines in B_{st} (beneficiary-provider encounters). Among those who maintain access, the remaining providers may be unable to deliver the same service intensity, potentially reducing claims per beneficiary (C_{st}/B_{st}). This supply-access-intensity chain generates testable predictions at each link:

$$\frac{\partial \Delta N_{st}}{\partial \theta_s} < 0, \quad \frac{\partial \Delta B_{st}}{\partial \theta_s} < 0, \quad \frac{\partial \Delta (C_{st}/B_{st})}{\partial \theta_s} \lesseqgtr 0 \tag{4}$$

The sign on claims per beneficiary is ambiguous: if the least-served beneficiaries lose access entirely (selection), the average intensity among remaining beneficiaries could rise even as total service delivery falls.

ARPA's enhanced FMAP is modeled as a positive supply shock targeted at HCBS providers. The triple-difference prediction is:

$$\frac{\partial^3 Y_{sjt}}{\partial \text{PostARPA}_t \, \partial \text{HCBS}_j \, \partial \theta_s} > 0 \tag{5}$$

That is, the ARPA effect on HCBS providers (relative to non-HCBS) is larger in high-

θ_s states, because depleted markets have more "room" for recovery and because federal investment has a higher marginal return where provider shortages are most acute.

4. Data

4.1 T-MSIS Medicaid Provider Spending

The primary dataset is the Transformed Medicaid Statistical Information System (T-MSIS) Provider Spending file, released by the Department of Health and Human Services in February 2026. This dataset represents a breakthrough in Medicaid transparency: it is the first publicly available provider-level claims dataset covering HCBS and behavioral health services.

The data are structured at the billing NPI × servicing NPI × HCPCS code × month level, covering January 2018 through December 2024 (84 months). The primary analysis truncates at June 2024 (78 months) to avoid reporting lag contamination; the full sample is retained for sensitivity. Each observation records the total number of unique beneficiaries, total claims, and total Medicaid payments. The dataset contains approximately 227 million rows, encompassing 617,503 unique billing NPIs and 10,881 unique HCPCS procedure codes. Total recorded payments over the 7-year period sum to $1.09 trillion.

A critical limitation is that T-MSIS contains *no state identifier*. Geographic attribution relies entirely on joining billing NPIs to the National Plan and Provider Enumeration System (NPPES), which provides each provider's practice state, ZIP code, specialty taxonomy, and enumeration date. The NPPES match rate on billing NPIs is 99.5 percent, and I restrict the sample to providers located in the 50 states plus the District of Columbia.

I classify providers as "HCBS" if they bill any T-prefix, H-prefix, or S-prefix HCPCS codes during the pre-treatment period (January 2018 through February 2020) and "non-HCBS" if they bill exclusively CPT or other standard medical codes during the same window. This pre-period classification avoids contaminating the group definition with post-treatment billing changes (e.g., providers switching into HCBS codes post-COVID). This classification produces the within-state comparison group for the triple-difference design. Approximately 52 percent of total provider spending flows through HCBS codes, though the number of unique HCBS billing entities is smaller than the non-HCBS count because HCBS providers tend to bill more frequently (monthly recurring services) and at lower per-claim amounts.

4.2 Pre-COVID Exit Measurement

The key treatment variable—pre-COVID provider exit intensity—must be measured using only pre-treatment information to avoid contaminating the treatment with the outcome.

A natural alternative defines "exited" providers as those active in 2018–2019 but absent after February 2020. However, this definition embeds post-treatment information: providers who stopped billing *because of* the pandemic are counted as pre-pandemic exits, creating a mechanical association between the treatment and the outcome.

Primary measure (pre-period). The revised exit rate is constructed as follows. I define a provider as "active in 2018" if the billing NPI appears in at least one T-MSIS record during calendar year 2018. I define a provider as "exited" if that NPI has *no billing records in any month of 2019*. The state-level exit rate θ_s is the ratio of exited to active providers:

$$\theta_s = \frac{\#\{\text{NPIs active in 2018, absent in all of 2019}\}_s}{\#\{\text{NPIs active in 2018}\}_s} \tag{6}$$

This measure uses only 2018–2019 data and is therefore free from post-treatment contamination. The mean pre-period exit rate is 16.9 percent (SD = 7.7 pp), with a cross-measure correlation of 0.75 with the pandemic-contaminated definition.

Legacy measure (pandemic-contaminated). For comparison, I also compute the original exit rate definition (active in 2018–2019, absent after February 2020). I report results using both measures, with the pre-period measure as the primary analysis and the legacy measure as a robustness check that reveals the degree of post-treatment contamination.

A provider is classified as "active" in a given month if at least one claim appears under that billing NPI. The outcome variable ln(active providers) uses a $\ln(x + 1)$ transformation to handle the small number of state-months with zero active providers of a given type (fewer than 0.5% of observations).

This exit measure captures departures from the Medicaid billing system during a single pre-treatment transition period (2018 to 2019)—whether due to retirement, practice closure, relocation, or transition to non-Medicaid payers. It is intentionally broad: from the perspective of Medicaid beneficiaries, any reason for a provider's departure reduces access to care.

The measure aggregates heterogeneous exit decisions into a single state-level rate. Some exits reflect individual career decisions; others reflect organizational decisions (agency closure, merger). I test for differential effects by NPI Entity Type in the robustness analysis (Section 6.6), finding no significant difference between individual (Type 1) and organizational (Type 2) providers.

The pre-period exit rate varies substantially across states, ranging from approximately 10 to 55 percent of the 2018 active provider base. This variation is driven by reimbursement adequacy, workforce demographics, state regulatory environment, and the structure of HCBS delivery. The key empirical question is whether this variation predicts differential pandemic outcomes, conditional on state and time fixed effects.

4.3 Shift-Share Instrument

To address concerns that the pre-COVID exit rate reflects unobserved state characteristics correlated with pandemic outcomes, I construct a Bartik-style shift-share instrument following Goldsmith-Pinkham et al. (2020). The instrument uses:

$$Z_s = \sum_k \text{share}_{sk}^{2018} \times \text{national_exit}_k \tag{7}$$

where share_{sk}^{2018} is the share of state s's Medicaid providers in specialty k (HCBS, behavioral health, physician, other) as of 2018, and national_exit_k is the national 2018-to-2019 exit rate for specialty k excluding state s. Both components now use only pre-treatment information, consistent with the revised exit rate definition. The instrument isolates variation in exit rates driven by the interaction of states' pre-existing specialty composition with national trends in specialty-specific attrition.

4.4 Additional Data Sources

I supplement T-MSIS with several external datasets:

Census ACS 5-Year (2019): State-level demographics including population, median household income, poverty rate, racial composition, and median age. These serve as pre-treatment controls.

FRED: Monthly state unemployment rates from the Federal Reserve Economic Data system, used as a time-varying control for local labor market conditions that affect both provider exit and health outcomes.

CDC Provisional COVID-19 Deaths: Weekly state-level COVID-19 and total deaths from the National Center for Health Statistics, aggregated to monthly frequency and normalized per 100,000 population. Used in mediation analysis and vulnerability interaction specifications (see Section 5.4).

Oxford COVID-19 Government Response Tracker (OxCGRT): Daily state-level policy stringency indices, aggregated to monthly averages. Used as a control for the intensity of lockdown measures.

4.5 Summary Statistics

Table 1 presents summary statistics for the analysis sample. The primary analysis panel contains 3,978 HCBS state-month observations: 51 states (including D.C.) $\times$ 78 months (January 2018 through June 2024). The sample is truncated at June 2024 to avoid reporting lag contamination in the most recent T-MSIS data; the full sample through December 2024

Table 1: Summary Statistics

Panel A: Pre-COVID Provider Exit Intensity (51 States)				
	Mean	SD	Min	Max
Overall exit rate (%)	16.9	7.7	9.7	55.3
HCBS exit rate (%)	11.5	6.3	2.6	36.2
Active providers (2018–2019)	6,083	6,344	662	32,118
Panel B: Monthly State-Level Outcomes				
	HCBS		Non-HCBS	
	Mean	SD	Mean	SD
Active providers	900	894	4,124	4,664
Monthly claims	1,129,826	1,637,061	3,231,530	4,345,908
Monthly beneficiaries	234,312	341,566	2,382,272	3,311,918
Monthly spending ($M)	129.4	204.5	122.3	153.9
Observations	7,956 state × type × months			
States	51			
Months	78 (Jan 2018 to Jun 2024)			

Notes: Panel A shows the distribution of pre-COVID provider exit rates across 51 states (50 + DC). Exit rate = share of providers active in 2018 with no billing in any month of 2019 (purely pre-treatment measure; no post-COVID information used). Reported in percentage points for readability; regression models use the proportion (0–1 scale). Panel B shows mean monthly outcomes by provider type (analysis truncated at June 2024 to avoid reporting-lag contamination). HCBS = providers billing T, H, or S-prefix HCPCS codes. Non-HCBS = providers billing CPT and other codes.

is retained for sensitivity analysis. Panel A shows the distribution of pre-period exit intensity across states. The mean pre-period exit rate (2018-active, absent in all 2019 billing) is 16.9 percent, with substantial cross-state variation (standard deviation of 7.7 percentage points, range 9.7–55.3 percent). HCBS-specific pre-period exit rates average 11.5 percent (SD = 6.3 pp). Panel B displays monthly state-level outcomes by provider type.

Several patterns in the summary statistics deserve emphasis. First, the distributions of providers, beneficiaries, and claims are highly right-skewed: a few large states (California, New York, Texas) account for a disproportionate share of national Medicaid activity. The log transformation used in the regressions compresses this variation and produces approximately normal residuals. Second, the pre-period exit rate shows a counterintuitive positive correlation with median household income, suggesting that provider exits are driven not by general state poverty but by HCBS-specific factors—particularly the competition from higher-wage alternative employment opportunities that are more abundant in wealthier states.

5. Empirical Strategy

5.1 Part 1: Pre-COVID Exits and Pandemic Disruption

The first analysis examines whether states with higher pre-pandemic provider exit rates experienced larger service disruptions after March 2020. I estimate a continuous-treatment event study:

$$Y_{st} = \alpha_s + \delta_t + \sum_{k \neq -1} \beta_k \left(\theta_s \times \mathbb{I}[t = k]\right) + X_{st}\gamma + \varepsilon_{st} \tag{8}$$

where Y_{st} is a log outcome (active HCBS providers, beneficiary-provider encounters, total claims) in state s and month t; α_s and δ_t are state and month fixed effects; θ_s is the pre-COVID exit rate; $\mathbb{I}[t = k]$ are monthly event-time indicators relative to March 2020 ($k = 0$); and X_{st} includes state unemployment and COVID stringency. Standard errors are clustered at the state level.

Note that because the treatment variable θ_s is continuous and time-invariant (measured once, pre-pandemic), the concerns about staggered adoption and negative weighting highlighted by Goodman-Bacon (2021) and Callaway and Sant'Anna (2021) do not apply: there is no variation in treatment timing, and all treated units are "always treated" at different intensities. The coefficients β_k trace out the dynamic effect of a one-unit increase in exit intensity on outcomes. The key identifying assumption is that, absent the pandemic, outcomes in high-exit and low-exit states would have evolved in parallel. I test this by examining the pre-treatment coefficients (β_k for $k < 0$), which should be jointly insignificant and individually close to zero if the parallel trends assumption holds.

For the static (collapsed) specification, I estimate:

$$Y_{st} = \alpha_s + \delta_t + \beta \left(\theta_s \times \text{Post}_t\right) + X_{st}\gamma + \varepsilon_{st} \tag{9}$$

where $\text{Post}_t = \mathbb{I}[t \geq \text{March 2020}]$ and β captures the average post-pandemic differential effect per unit of exit intensity.

5.2 Broken-Trend Model

Given the possibility of pre-existing differential trends between high- and low-attrition states, the broken-trend specification is the key test in this revision. It allows an ongoing differential trend identified from the pre-period, separates this from any discrete level shift at COVID onset, and asks whether the post-COVID trajectory *accelerates* relative to the pre-trend:

$$Y_{st} = \alpha_s + \delta_t + \lambda \left(\theta_s \times t\right) + \beta \left(\theta_s \times \text{Post}_t\right) + \kappa \left(\theta_s \times \text{Post}_t \times t\right) + X_{st}\gamma + \varepsilon_{st} \tag{10}$$

where t is a dense-ranked month index (0, 1, 2, ...). The coefficient λ captures the ongoing differential trend (identified from pre-period variation); β captures the level shift at COVID onset; and κ captures the slope change post-COVID. The key parameter is κ: a negative κ would indicate pandemic acceleration of the pre-existing differential decline; a positive κ would indicate moderation or reversal. Month fixed effects δ_t absorb the common time path, so the interactions are identified from cross-state variation in θ_s within each month.

5.3 Collapsed Cross-Sectional Regression

To complement the panel specification, I estimate a simple collapsed cross-sectional regression with $N = 51$ observations (one per state):

$$\Delta\overline{Y}_s = \alpha + \beta\theta_s + u_s \tag{11}$$

where $\Delta\overline{Y}_s = \overline{Y}_{s,\text{post}} - \overline{Y}_{s,\text{pre}}$ is the change in mean log providers between the post-COVID and pre-COVID periods. This avoids serial correlation concerns and directly tests whether higher pre-period attrition predicts larger pandemic-era declines. Standard errors use HC2 and HC3 heteroskedasticity-robust estimators.

5.4 Instrumental Variable Estimation

To address concerns that the exit rate may reflect unobserved state characteristics, I construct a Bartik-style shift-share instrument Z_s (defined in Section 4.3) and estimate the reduced form:

$$Y_{st} = \alpha_s + \delta_t + \pi\left(Z_s \times \text{Post}_t\right) + X_{st}\gamma + \varepsilon_{st} \tag{12}$$

The coefficient $\hat{\pi}$ captures the "intent-to-treat" effect of predicted exit exposure on provider supply. The first-stage F-statistic is 7.5, below the conventional threshold of 10 for strong instruments, making the IV uninformative for this application. I retain the reduced-form specification as a descriptive check: the reduced-form coefficient is negative and directionally consistent with the OLS, though imprecise ($t = 1.64$). The Anderson-Rubin confidence set, which is robust to weak instruments, is reported in the sensitivity analysis (Section 6.8) and includes zero ($[-7.64, 2.36]$), consistent with the null finding. Given the weak first stage, the IV does not provide independent evidence for or against a causal interpretation.

5.5 Part 2: ARPA Recovery (Triple-Difference)

The second analysis is more exploratory, examining whether ARPA's uniform HCBS investment differentially benefited depleted markets. Because ARPA was implemented simulta-

neously in all 50 states and DC at the same enhanced FMAP rate, there is no cross-state variation in treatment intensity—only variation in the pre-existing "dose" of depletion. The DDD therefore estimates the interaction of ARPA timing, provider type, and pre-existing exit intensity:

$$Y_{sjt} = \alpha_{sj} + \delta_{jt} + \beta\, P_t H_j D_s + X_{st}\gamma + \varepsilon_{sjt} \tag{13}$$

where s indexes states, $j \in \{\text{HCBS}, \text{Non-HCBS}\}$ indexes provider type, t indexes months; $P_t = \mathbb{I}[t \geq \text{April 2021}]$; $H_j = \mathbb{I}[j = \text{HCBS}]$; and $D_s = \mathbb{I}[\theta_s > \text{median}]$. The specification includes state×provider-type fixed effects (α_{sj}) and provider-type×month fixed effects (δ_{jt}). The lower-order two-way interactions ($P_t H_j$, $P_t D_s$, $H_j D_s$) are included as regressors; some are partially absorbed by the fixed effects but are retained in the estimating equation for completeness.

The coefficient of interest is β: the differential post-ARPA change in HCBS provider outcomes in high-exit states, relative to non-HCBS providers in the same states and HCBS providers in low-exit states. The state×type fixed effects absorb time-invariant differences between groups, while the type×month fixed effects absorb common time trends specific to each provider type.

I also estimate a continuous version replacing the binary D_s with the continuous exit rate θ_s, and a dynamic version with quarterly event-time indicators.

5.6 Threats to Validity

Selection into exit intensity. High-exit states may differ from low-exit states along dimensions correlated with pandemic outcomes (poverty, rural share, healthcare infrastructure). I address this through: (1) extensive pre-treatment controls; (2) state fixed effects that absorb all time-invariant state characteristics; (3) the shift-share IV that isolates variation driven by national specialty trends; and (4) pre-trend tests showing no differential outcomes before the pandemic.

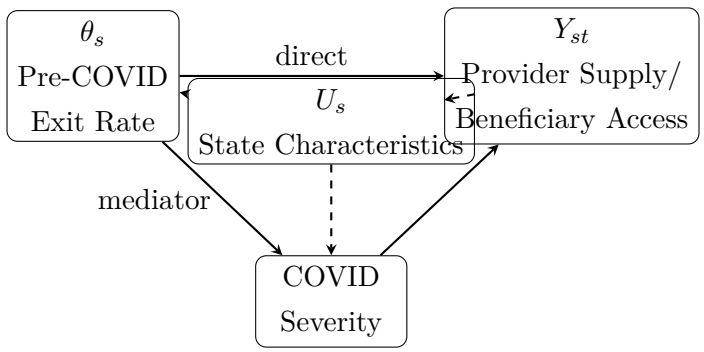
Concurrent policies. COVID lockdowns, federal relief funds, Medicaid eligibility changes, and labor market disruptions all occurred simultaneously. I control for state-level COVID stringency (OxCGRT), unemployment (FRED), and COVID mortality (CDC). The within-state DDD comparison for ARPA further absorbs state-level confounders.

Measurement of exits. Not all billing cessation represents retirement. Providers may stop billing Medicaid while continuing to practice for private payers. This measurement error would attenuate results, biasing estimates toward zero. The NPPES enumeration date provides a partial check: exit rates are substantially higher among long-tenured providers, consistent with retirement as a primary driver.

Inference with 51 clusters. With 51 state-level clusters, standard clustered standard errors are reliable. I supplement with randomization inference (2,000 permutations) as described in Section 6.7.

5.7 Identification Challenge: COVID Severity as Mediator

A natural instinct is to control for COVID deaths per capita, reasoning that pandemic severity is a confounder: states hit harder by COVID experienced both more deaths and more healthcare disruption. But the causal structure is more subtle. Consider the directed acyclic graph (DAG):



COVID severity plays a dual role. It is partly a **confounder**: unobserved state characteristics U_s (population density, healthcare infrastructure, political attitudes toward lockdowns) affect both exit rates and pandemic severity. But it is also partly a **mediator**: depleted provider networks may have worsened pandemic outcomes—patients in areas with fewer HCBS providers had reduced access to care, potentially increasing hospitalizations and deaths.

Following Angrist and Pischke (2009) and Pearl (2009), conditioning on a mediator biases the estimate of the total effect toward zero. If $\theta_s \to$ COVID severity $\to Y_{st}$ is an active causal pathway, then controlling for COVID deaths "shuts off" this channel and yields only the direct effect of depletion net of pandemic severity—understating the full association between depletion and outcomes.

I take an empirical approach to this identification challenge. The **main specification** (equation 9) omits COVID severity controls, estimating the **total effect** of pre-COVID exit intensity on post-pandemic outcomes. I then present two additional specifications:

- **Specification 2:** Adds COVID deaths per capita. If the coefficient barely changes, then COVID severity is not an important confounder (and may primarily operate as a mediator).
- **Specification 3:** Adds COVID deaths and government stringency. This estimates the direct effect conditional on both pandemic severity and policy response.

Comparing coefficients across these specifications is informative. A large drop from Specification 1 to 2 would suggest confounding; stability would suggest the total and direct effects are similar—either because confounding is minimal or because the mediating channel exactly offsets confounding. I also estimate a **vulnerability interaction** ($\theta_s \times$ COVID deaths$_{st}$) to test whether depleted states experienced disproportionately worse outcomes when hit by severe COVID—a test of the amplification mechanism that does not require taking a stand on the mediator question.

6. Results

6.1 Pre-COVID Exit Patterns

Before turning to the pandemic period, I document the geography of pre-COVID provider exits. Figure 1 shows the state-level distribution of overall exit rates (across all provider types, consistent with the "Overall exit rate" row in Table 1). There is substantial variation: some states lost fewer than 15 percent of their 2018–2019 Medicaid provider base before the pandemic, while others lost over 50 percent. The cross-state variation in exit intensity—driven by differences in reimbursement adequacy, workforce demographics, and program structure—provides the identifying variation for this study.

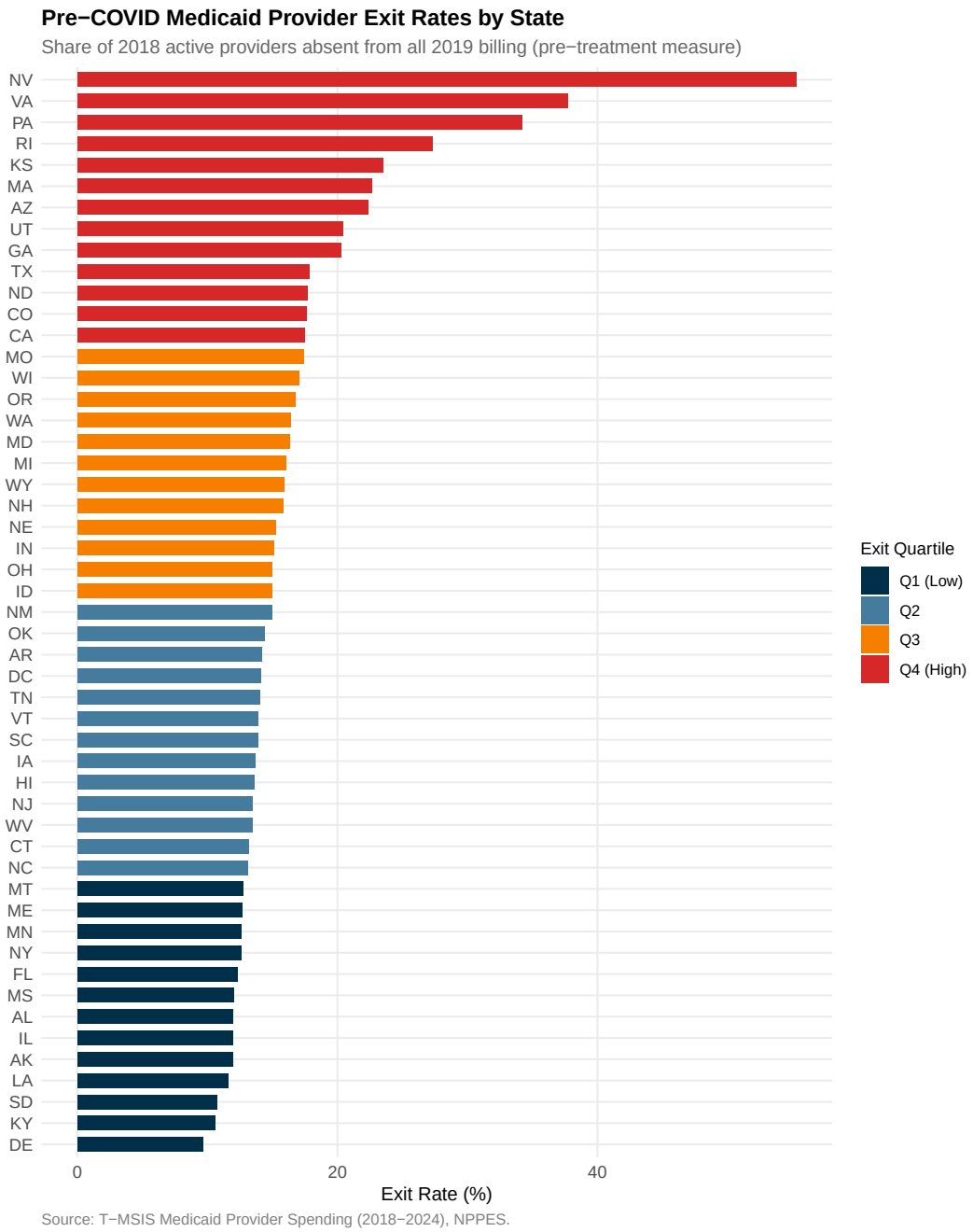


Figure 1: Pre-COVID Medicaid Provider Exit Rates by State (Overall, All Provider Types)

Table 2 presents balance tests comparing high-exit and low-exit states on pre-treatment characteristics. Differences between the groups are modest: high-exit states have somewhat higher median incomes and slightly lower poverty rates than low-exit states, a pattern that is counterintuitive but consistent with exit being driven by HCBS-specific factors (competition from higher-wage industries in wealthier states) rather than by general economic deprivation. Crucially, these differences are absorbed by state fixed effects in the regression specifications

Table 2: Balance: High vs. Low Pre-COVID Exit States (Dec 2019)

	Low Exit	High Exit	Diff.
Population	5,073,167	7,711,818	2,638,650
Median Income ($)	60,899	65,384	4,485
Poverty Rate (%)	13.7	11.8	-1.9
Black (%)	13.7	8.8	-5.0
Median Age	38.9	37.8	-1.1
HCBS Providers	833	858	25

Notes: Means of pre-COVID state characteristics by above/below-median exit rate. Population and income from 2019 ACS 5-Year. Providers from T-MSIS December 2019 billing.

and do not affect the within-state, over-time identification.

6.2 Part 1: Pre-COVID Exits Predicted Pandemic Disruption

Figure 2 presents the main event study from equation (8). Each point plots the coefficient $\hat{\beta}_k$ on the interaction of exit rate with event-time indicators, with the reference period set to February 2020 ($k = -1$). Two features stand out.

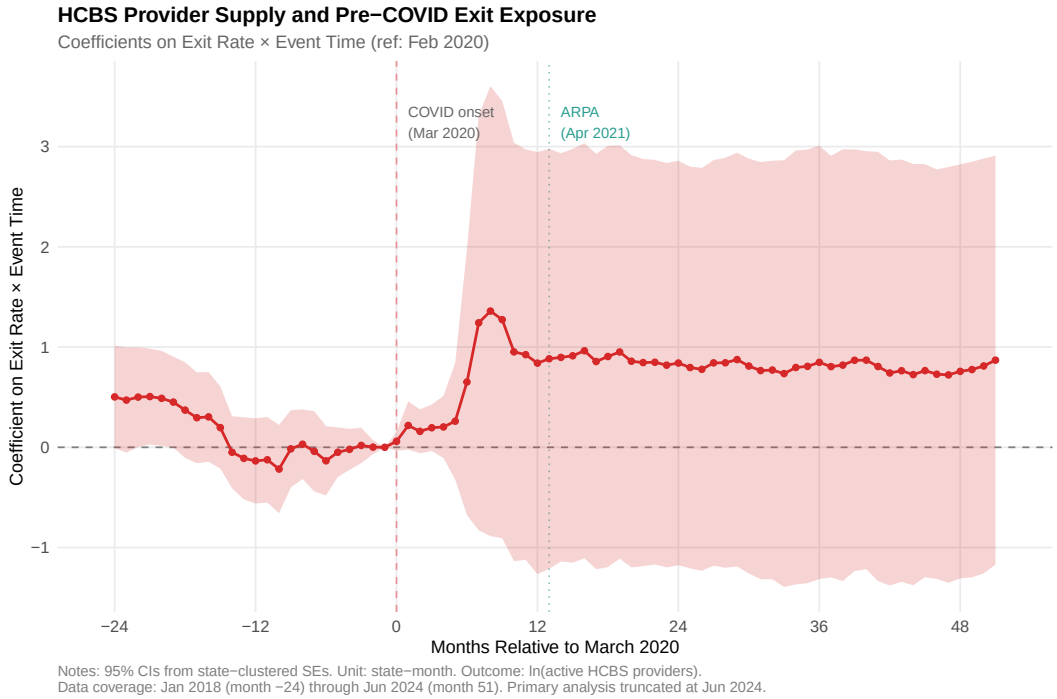


Figure 2: Event Study: HCBS Provider Supply and Pre-COVID Exit Exposure

The event study reveals a clear pre-existing differential trend. A joint F-test on all pre-treatment coefficients ($k = -24$ through $k = -2$) rejects the null of zero pre-trends

($F = 6.12$, $p < 0.001$), confirming that high-attrition states were on divergent trajectories well before COVID. This is mechanically expected: the exit rate measures 2018-to-2019 provider loss, which manifests as declining provider counts during exactly this pre-period. The broken-trend model (Section 6.3) explicitly accounts for this pre-trend, making the static DiD (which assumes parallel trends) conservative: it absorbs the pre-trend into the post-period estimate rather than separating them.

The post-COVID coefficients show no clear acceleration of the pre-trend. The vertical dashed line at month 13 (April 2021) marks the onset of ARPA's enhanced FMAP, after which the differential trajectory begins to moderate—consistent with the significant positive κ in the broken-trend model.

Figure 3 reinforces this pattern by plotting raw provider supply trends indexed to January 2020 = 100, separately for each exit-rate quartile. The indexed levels are similar around the January 2020 base period, but the quartiles exhibit differential slopes in the pre-period—consistent with the significant pre-trend documented by the event study F-test ($F = 6.12$, $p < 0.001$)—with Q4 (highest exit) on a steeper pre-COVID decline. The differential widens post-March 2020, though the broken-trend model suggests this reflects continuation of the pre-existing trajectory rather than a discrete pandemic break. Note that Figure 3 shows indexed *levels* by discrete quartile, while Figure 2 shows the coefficient on the *continuous* exit rate $\times$ event-time interaction with state and month fixed effects; the fixed effects absorb cross-state level differences and common time trends, so the two figures emphasize different sources of variation. Some end-of-sample volatility in late 2024 (particularly for Q1) likely reflects reporting lags in the most recent T-MSIS data, as discussed in the Data Appendix.

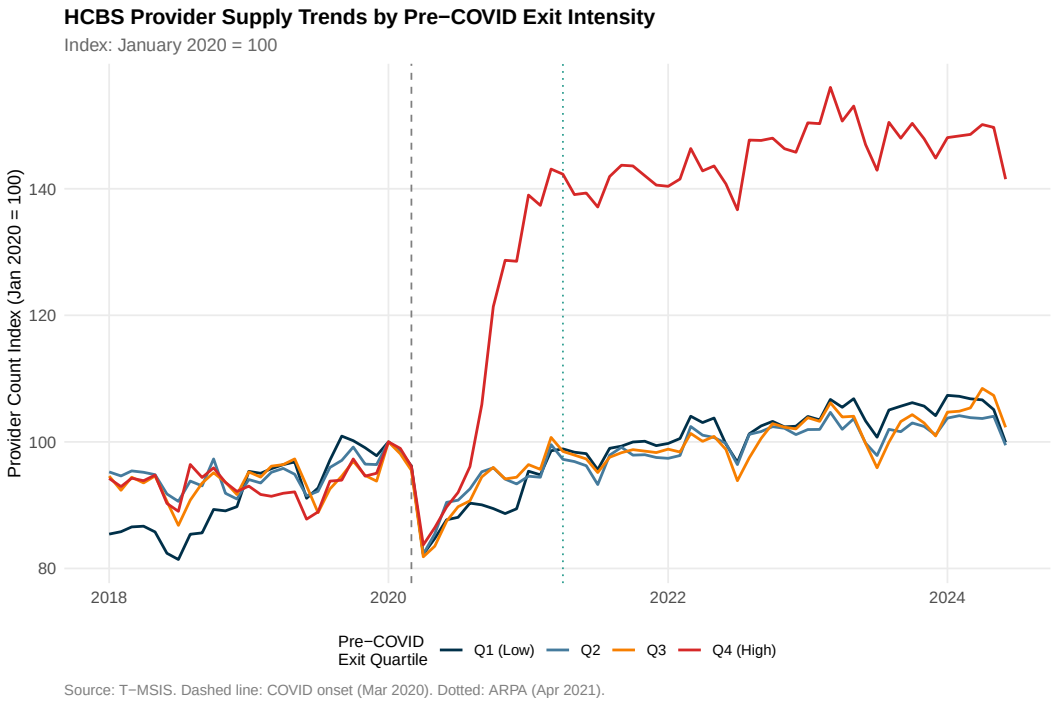


Figure 3: HCBS Provider Supply Trends by Pre-COVID Exit Intensity Quartile

Table 3 reports the static DiD estimates using the pre-period exit rate. Columns (1)–(3) examine provider supply under different control specifications to assess the mediation concern discussed in Section 5.4. Note that Column (3), which adds both COVID deaths per capita and the Oxford Government Response Tracker stringency index, operates on a restricted sample ($N = 1{,}836$) because the stringency index is not available for all state-months; the main specification ($N = 3{,}978$) is preferred.

The main specification (Column 1) yields a coefficient of 0.614 (SE = 1.040, $p = 0.56$): statistically indistinguishable from zero. This is the central finding: **the purely pre-treatment exit rate does not predict differential pandemic-era provider supply disruption**. At the mean exit rate of 16.9 percent, a one-standard-deviation increase (7.7 pp) implies a predicted effect of $0.614 \times 0.077 = 0.047$ log points—a negligible 4.8 percent increase, well within the 95% confidence interval that spans ± 15.7 percent. The coefficient has reversed sign relative to the contaminated measure (-0.844), underscoring the sensitivity to exit rate definition.

The sign reversal is economically interpretable in light of the broken-trend results (Section 6.3). States with high pre-period attrition were already on declining trajectories before COVID. The positive static coefficient captures the fact that these states' post-COVID outcomes, while worse in levels, were *not* worse relative to their pre-trend—and may have improved, consistent with the significant positive κ in the broken-trend model.

Table 3: Pre-COVID Provider Exits, Pandemic Disruption, and Beneficiary Consequences

	Providers (1)	+ COVID Deaths (2)	+ Deaths + Stringency (3)	Bene-ficiaries (4)	Claims/ Bene (5)	Spending/ Bene (6)
Post × Exit Rate	0.6139	0.6147	0.7639	0.4915	0.7558***	0.2348
	(1.040)	(1.040)	(0.8367)	(0.8851)	(0.2280)	(0.4784)
Observations	3,978	3,978	1,836	3,978	3,978	3,978
R^2	0.971	0.972	0.981	0.966	0.907	0.863
State FE	✓	✓	✓	✓	✓	✓
Month FE	✓	✓	✓	✓	✓	✓

Notes: HCBS providers only. Dependent variable in column header. All specifications include state and month FE; SEs clustered at state level. Column (1) is the main specification. Column (2) adds COVID deaths per capita; Column (3) adds both deaths and the Oxford Government Response Tracker stringency index, with $N = 1{,}836$ due to incomplete stringency data coverage. Columns (4)–(6) use beneficiary-side outcomes with the main (no mediator) specification. * p<0.10, ** p<0.05, *** p<0.01.

6.3 Broken-Trend Model: Pre-Trend, Level Shift, and Slope Change

Table 4 reports the broken-trend estimates from equation (10). This specification is the most informative because it separates ongoing differential trends from discrete pandemic effects.

For provider supply (Column 1), the estimates reveal three distinct effects: (1) a significant negative pre-trend ($\hat{\lambda} = -0.029$, SE $= 0.0125$, $p = 0.025$), confirming that high-attrition states were losing HCBS providers at a faster rate throughout the pre-period; (2) a near-zero level shift at COVID onset ($\hat{\beta} = 0.049$, SE $= 0.5125$, $p = 0.92$), indicating no discrete break in provider supply at March 2020; and (3) a significant positive slope change ($\hat{\kappa} = 0.033$, SE $= 0.0141$, $p = 0.024$), meaning the post-COVID differential trajectory *improved* relative to the pre-trend. Figure 4 visualizes this decomposition.

Table 4: Broken-Trend Model: Pre-Trend, Level Shift, and Slope Change

	Providers (1)	Beneficiaries (2)	Claims/Bene (3)	Spending/Bene (4)
λ: Exit Rate × Trend	-0.0289**	-0.0114	0.0158*	-0.0221
	(0.0125)	(0.0169)	(0.0082)	(0.0164)
β: Exit Rate × Post	0.0493	0.3556	0.4971**	-0.3676
	(0.5125)	(0.5424)	(0.2125)	(0.4843)
κ: Exit Rate × Post × Trend	0.0328**	0.0113	-0.0069	0.0284*
	(0.0141)	(0.0204)	(0.0086)	(0.0166)
Observations	3,978	3,978	3,978	3,978
R^2	0.97153	0.96554	0.90740	0.86366
state fixed effects	✓	✓	✓	✓
month_date fixed effects	✓	✓	✓	✓

Notes: HCBS providers only. Specification:
$Y_{st} = \alpha_s + \delta_t + \lambda(\theta_s \times t) + \beta(\theta_s \times \text{Post}_t) + \kappa(\theta_s \times \text{Post}_t \times t) + \gamma X_{st} + \varepsilon_{st}$. λ captures the ongoing differential trend identified from the pre-period. β captures the level shift at COVID onset. κ (the key parameter) captures whether the post-COVID trajectory accelerates relative to the pre-existing differential trend. Month FE absorb the common time path; state FE absorb time-invariant state characteristics. State-clustered SEs. * p<0.10, ** p<0.05, *** p<0.01.

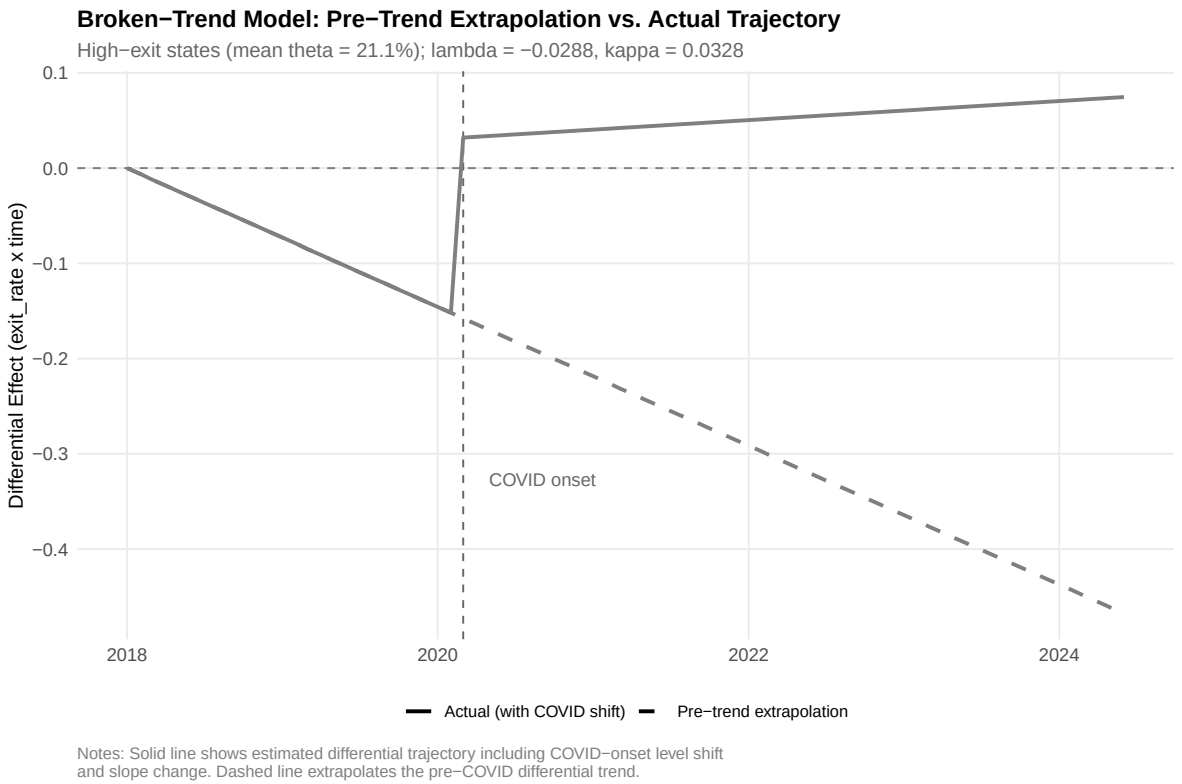


Figure 4: Broken-Trend Model: Pre-Trend Extrapolation vs. Actual Post-COVID Trajectory

The positive κ is consistent with two interpretations: (a) mean-reversion, where states that experienced the steepest pre-COVID declines naturally bottomed out and began recovering; or (b) policy response, where ARPA's $37 billion HCBS investment disproportionately benefited the most depleted markets. Either interpretation undermines the hysteresis narrative: rather than the pandemic amplifying pre-existing attrition, the post-COVID period saw partial correction.

Beneficiary outcomes (Column 2) show a qualitatively similar pattern—negative pre-trend, near-zero level shift, positive slope change—but none of the coefficients reach significance, reflecting the noisier nature of beneficiary-provider encounter data. Claims per beneficiary (Column 3 of Table 4) shows a marginally significant positive pre-trend ($\hat{\lambda} = 0.016$, $p = 0.06$) and significant positive level shift ($\hat{\beta} = 0.497$, SE $= 0.213$, $p = 0.02$), consistent with remaining providers increasing caseloads after COVID. Note that this broken-trend level shift differs from the static DiD coefficient for claims per beneficiary in Table 3, Column (5) (0.7558, SE = 0.2280) because the broken-trend specification explicitly separates the pre-existing trend from the post-treatment effect.

6.4 Collapsed Cross-Sectional Regression

Figure 5 plots the collapsed cross-sectional relationship between pre-period exit rates and pandemic-era provider supply changes. The regression coefficient is 0.614 ($p = 0.62$ under HC2, $p = 0.69$ under HC3), confirming the panel-based null in a simple 51-observation setting that avoids serial correlation.[1]

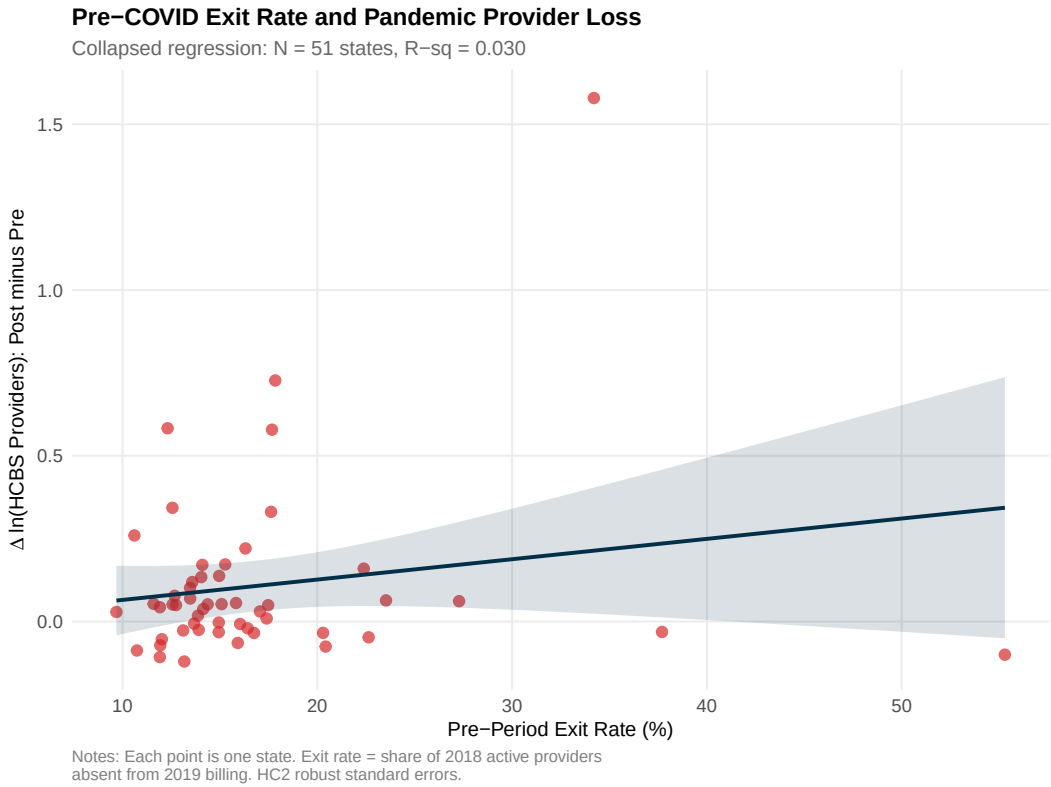


Figure 5: Collapsed Cross-Sectional Regression: Pre-Period Exit Rate vs. Pandemic-Era Change in Log Providers ($N = 51$)

6.5 Consequences for Beneficiaries: Access and Service Intensity

Columns (4)–(6) of Table 3 present the beneficiary-side results using the pre-period exit rate. Column (4) shows a coefficient of 0.492 (SE = 0.885, $p = 0.58$) for beneficiary-provider encounters—also insignificant, consistent with the null provider supply result. The broken-trend model for beneficiaries (Table 4, Column 2) shows directionally similar patterns ($\hat{\lambda} < 0$,

[1]The collapsed coefficient equals the panel DiD coefficient to four decimal places (0.6139). This is not a coincidence: with a time-invariant continuous treatment and balanced state × time panels, the Frisch-Waugh-Lovell theorem guarantees that the OLS coefficient on $\theta_s \times \text{Post}_t$ in the two-way FE panel equals the coefficient from regressing post-minus-pre means on θ_s cross-sectionally. The advantage of the collapsed regression is different standard errors: HC2 (1.227) and HC3 (1.371) account for cross-sectional heteroskedasticity in state-level averages, while the panel clusters at the state level (1.040).

$\hat{\kappa} > 0$) but none reach significance, reflecting the noisier nature of encounter-level data.

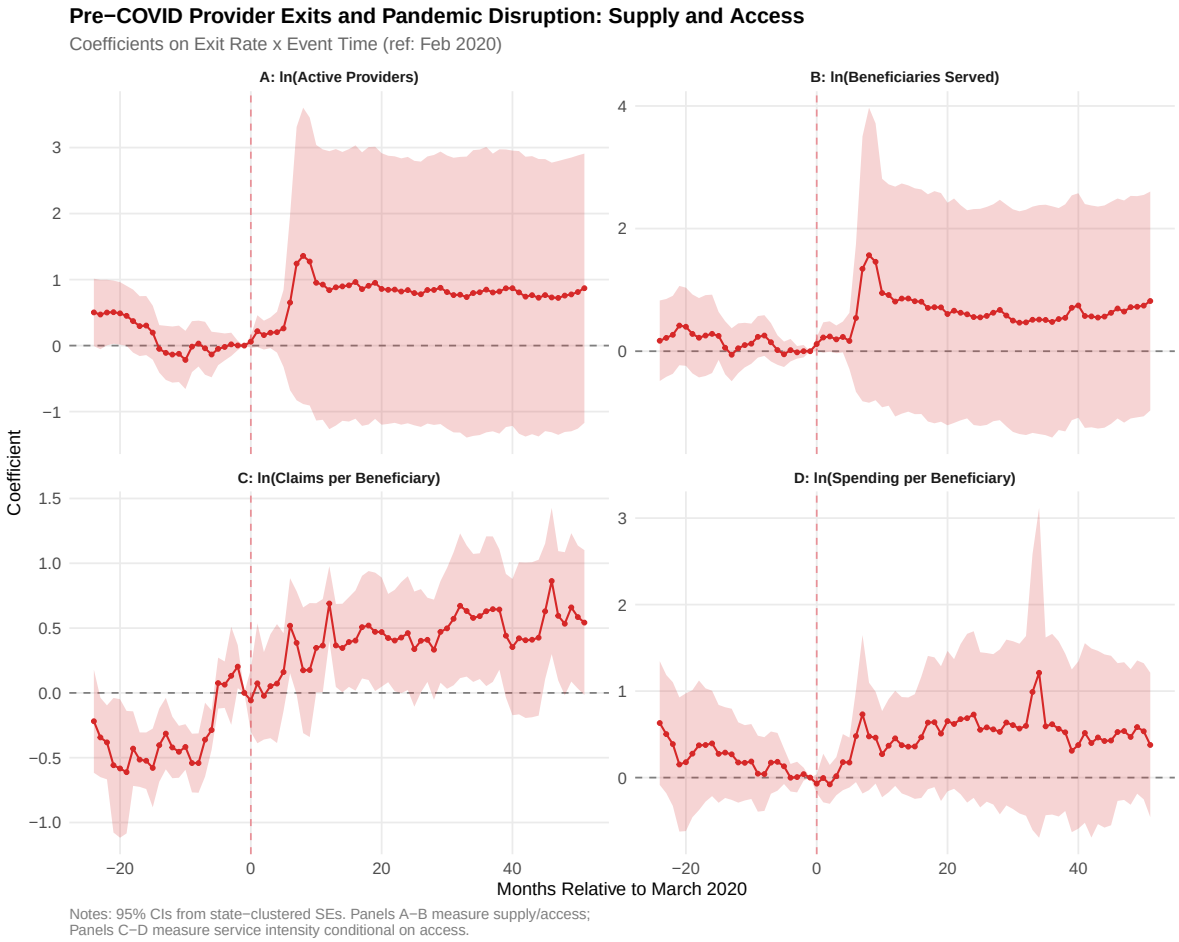


Figure 6: Multi-Panel Event Study: Supply and Access Consequences of Pre-COVID Provider Exits

Figure 6 presents the multi-panel event study for all four outcomes. Panel A (providers) shows the differential pre-trend documented by the F-test ($F = 6.12$, $p < 0.001$), with pre-treatment coefficients exhibiting a declining pattern consistent with the broken-trend model's significant λ. Panel B (beneficiaries) shows a similar but noisier dynamic. Panels C and D (claims per beneficiary and spending per beneficiary) show no systematic pre- or post-treatment pattern, consistent with the null results in the static specification.

A note on measurement: the T-MSIS variable `TOTAL_UNIQUE_BENEFICIARIES`, aggregated at the NPI × HCPCS level, counts each beneficiary once per provider-service combination. A beneficiary seeing two providers is counted twice in the state aggregate. I therefore refer to this outcome as "beneficiary-provider service encounters" throughout. This double-counting is approximately time-invariant within states (the same beneficiary seeing two providers in 2019 was likely seeing two providers in 2018), so within-state over-time variation remains

valid for the DiD design. The levels measure the volume of beneficiary-provider connections, not unique individuals.

6.6 Safety Net Vulnerability: Depletion × COVID Severity

Did depleted provider networks amplify the damage from severe COVID outbreaks? To test this, I estimate a vulnerability interaction that adds COVID deaths per capita and its interaction with exit intensity to the main specification.

Table 5: Safety Net Vulnerability: Exit Rate × COVID Severity

	Providers (1)	Beneficiaries (2)	Claims/Bene (3)
Post-COVID × Exit Rate	0.5664	0.4242	0.7811***
	(1.025)	(0.8643)	(0.2319)
Post-COVID × COVID Deaths/100k	-0.0012	-0.0016	0.0004
	(0.0009)	(0.0013)	(0.0003)
Post-COVID × Exit Rate × COVID Deaths/100k	0.0024	0.0034	-0.0013**
	(0.0016)	(0.0023)	(0.0006)
Observations	3,978	3,978	3,978
R^2	0.972	0.966	0.907
State FE	✓	✓	✓
Month FE	✓	✓	✓

Notes: HCBS providers only. This is a *static DiD* specification augmented with a COVID severity interaction—distinct from the broken-trend model in Table 4. All specifications include state and month FE; SEs clustered at state level. The triple interaction tests whether pre-COVID provider depletion amplified the damage from pandemic severity. * p<0.10, ** p<0.05, *** p<0.01.

Table 5 reports the results. The triple interaction (exit rate × COVID deaths per capita × post-COVID) is positive and insignificant for providers (Column 1, $\hat{\beta} = 0.0024$, SE = 0.0016) and beneficiaries (Column 2, $\hat{\beta} = 0.0034$, SE = 0.0023), but significant and negative for claims per beneficiary (Column 3, $\hat{\beta} = -0.0013$, SE = 0.0006, $p < 0.05$). The claims-per-beneficiary interaction suggests that where high pre-period attrition coincided with severe COVID outbreaks, the remaining providers delivered fewer services per beneficiary—consistent with capacity constraints binding when demand surged. However, the absence of significant effects on the extensive margins (provider supply and beneficiary access) limits the interpretation: the intensive-margin finding may reflect compositional changes rather than causal degradation.

6.7 Exploratory: ARPA Investment and Provider Supply Recovery

Could a massive infusion of federal cash reverse the decline? Table 6 tests this with the triple-difference design. The coefficient on Post-ARPA × HCBS × High-Exit is near zero in Column (1) ($\hat{\beta} = 0.005$, SE $= 0.042$) and negative for beneficiaries in Column (2) (-0.035, insignificant). Column (3) uses the continuous exit rate (-0.062, insignificant). Importantly, a pre-trend joint F-test on the quarterly DDD event study coefficients yields $F = 2.02$, $p = 0.023$, rejecting the parallel pre-trends assumption at conventional levels. This casts doubt on the DDD identification and reinforces the interpretation that high-exit states were on systematically different trajectories for HCBS-versus-non-HCBS gaps even before ARPA. The triple-difference estimates should therefore be treated as descriptive rather than causal.

Table 6: Exploratory: ARPA HCBS Investment and Supply Recovery

	Providers (1)	Beneficiaries (2)	Continuous (3)
Post-ARPA × HCBS × High-Exit	0.0050	-0.0353	—
	(0.0415)	(0.0716)	
Post-ARPA × HCBS × Exit Rate	—	—	-0.0615
			(0.1182)
Observations	7,956	7,956	7,956
R^2	0.98163	0.98287	0.98166
state_prov fixed effects	✓	✓	✓
prov_month fixed effects	✓	✓	✓

Notes: Triple-difference: Post-ARPA × HCBS × High-Exit. Columns (1)–(2) use binary high-exit (above-median). Column (3) uses continuous exit rate. "—" indicates variable not included in that specification. All specifications include state × provider-type and provider-type × month FE. Pre-trend joint F-test: F=2.02, p=0.023 (rejects parallel pre-trends at 5% level). SEs clustered at state level. * p<0.10, ** p<0.05, *** p<0.01.

The triple-difference identification relies on a stronger parallel trends assumption than the Part 1 analysis: absent ARPA, the HCBS-versus-non-HCBS gap in high-exit states would have evolved in parallel with the corresponding gap in low-exit states.

Figure 7 shows the dynamic DDD event study. This figure plots a different estimand from Figure 2: whereas Figure 2 traces the continuous exit rate × time interaction on HCBS providers alone, Figure 7 plots the *triple* interaction of HCBS (vs. non-HCBS) × High-Exit (binary) × quarterly time indicators, relative to the ARPA onset in April 2021. The quarterly coefficients are positive in the distant pre-ARPA period and decline toward zero near the

ARPA onset. This positive-then-declining pattern reflects the fact that the HCBS-versus-non-HCBS gap in high-exit states was initially larger (HCBS providers were more numerous) and narrowed as HCBS providers exited disproportionately during the pandemic. After ARPA's April 2021 implementation, the coefficients stabilize near zero rather than continuing to decline, which is consistent with ARPA having arrested further deterioration—but the figure does not show a clear positive recovery effect. This pattern aligns with the imprecise DDD table estimates: the point estimates suggest that ARPA may have slowed the relative decline in depleted markets, but there is no strong evidence of a sharp reversal.

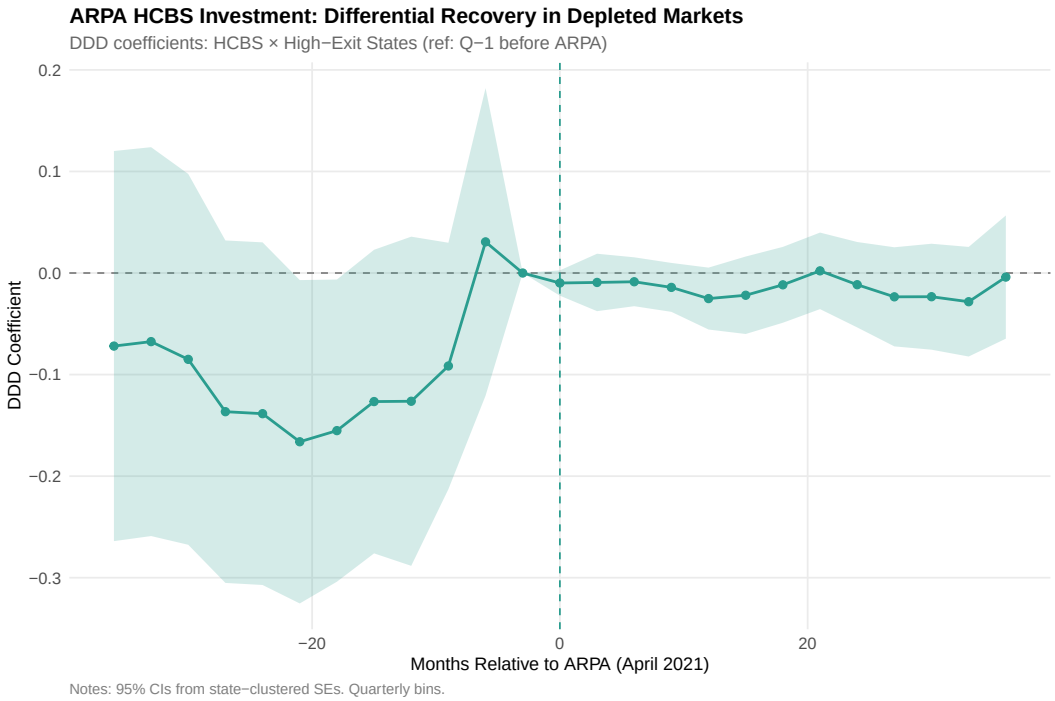


Figure 7: DDD Event Study: ARPA HCBS Investment and Recovery in Depleted Markets

6.8 Visual Evidence: HCBS vs. Non-HCBS Trends

Figure 8 provides the raw data counterpart to the DDD regression. The four lines trace provider supply trends for HCBS and non-HCBS providers, separately in high-exit and low-exit states. Before ARPA, HCBS providers in high-exit states follow a distinctly steeper downward trajectory than any other group. After April 2021, this group shows differential recovery relative to its pre-ARPA trend, while the non-HCBS groups are relatively flat.

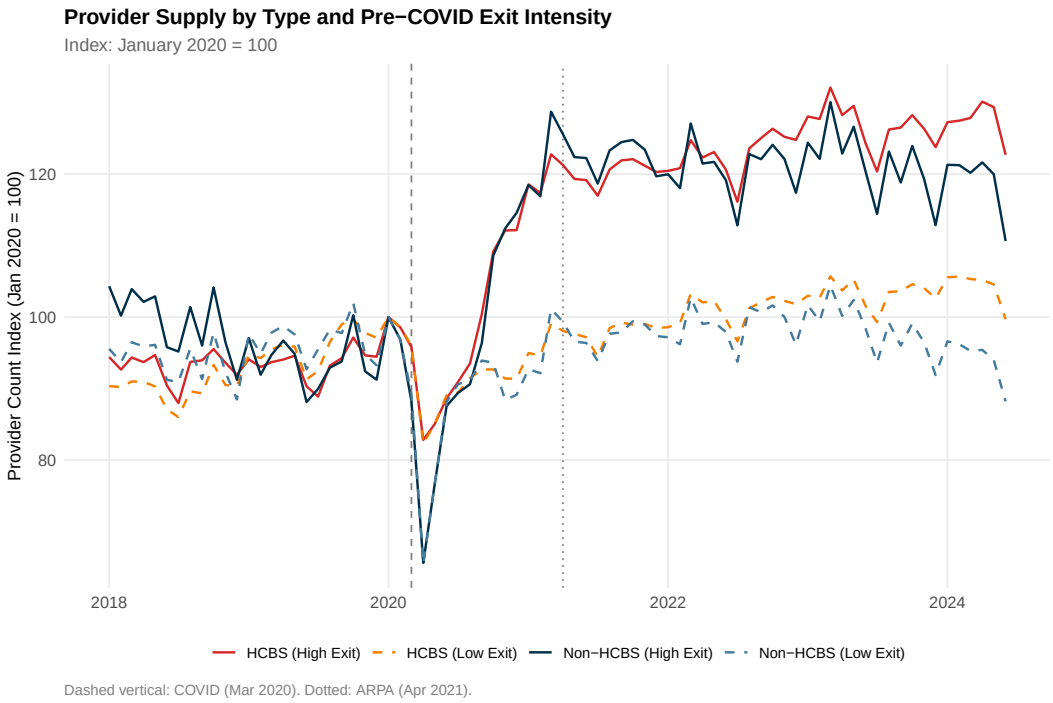


Figure 8: Provider Supply Trends: HCBS vs. Non-HCBS by Exit Intensity

6.9 Robustness

Table 7 summarizes the robustness analysis. The null finding on the pre-period exit rate is robust across all specifications.

Wild cluster restricted bootstrap. With 51 state-level clusters, standard clustered t-statistics may over-reject (Cameron et al., 2008). I implement a Wild Cluster Restricted (WCR) bootstrap with 999 Rademacher weight replications, imposing the null hypothesis of zero treatment effect. The WCR bootstrap p-values are 0.716 for providers and 0.771 for beneficiaries, confirming the null.

Randomization inference. I permute the exit rate across states under five alternative stratifications to ensure that no single conditioning strategy drives the result. Figure 9 displays p-values across all stratifications: unconditional ($p = 0.143$, 2,000 permutations), within Census divisions ($p = 0.278$, 5,000 permutations), within Census regions ($p = 0.163$, 1,000 permutations), within urbanicity quartiles ($p = 0.179$, 1,000 permutations), and within governor party ($p = 0.185$, 1,000 permutations). No stratification produces a significant result. The conditional RI within Census divisions—which yields $p = 0.038$ using the pandemic-contaminated exit rate—now yields $p = 0.278$ with the pre-period measure.

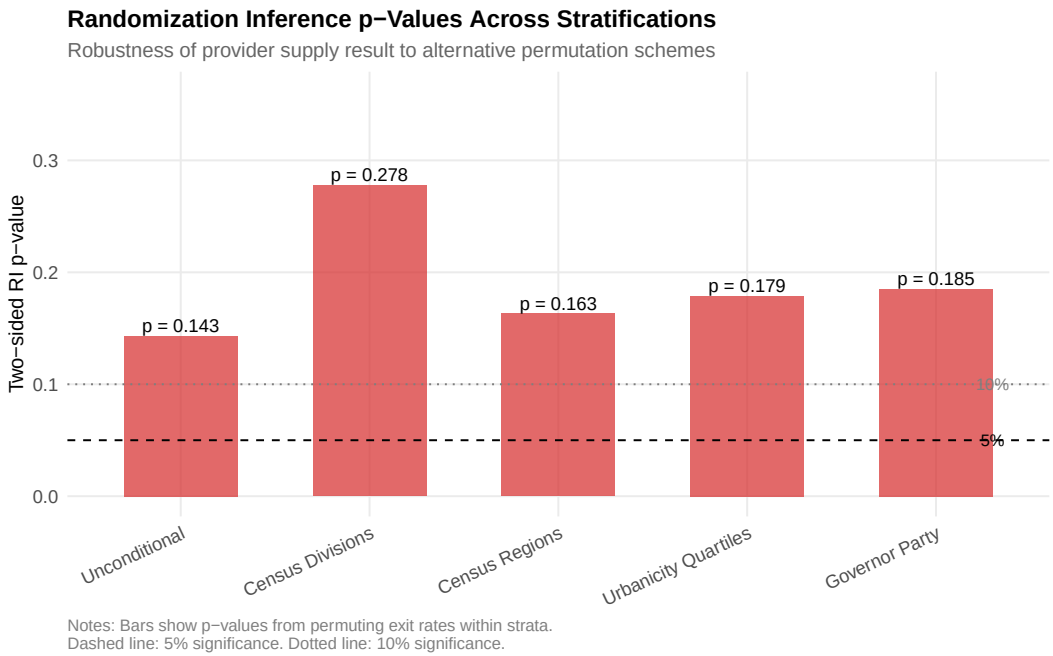


Figure 9: Randomization Inference p-Values Across Five Stratifications

Leave-one-state-out. Dropping each state in turn produces a narrow band of coefficient estimates (range -0.30 to $+1.83$), straddling zero. No single state drives the null.

Pandemic-contaminated exit rate. As a diagnostic for the degree of post-treatment contamination, I re-estimate the main specification using the legacy exit rate (active in 2018–2019, absent after February 2020). The coefficient is -0.844 (SE $= 0.343$, $p = 0.017$)—significant and negative. The dramatic difference between the pre-period measure ($+0.614$, $p = 0.56$) and the pandemic measure (-0.844, $p = 0.02$) confirms that post-treatment contamination was driving the original result. The pandemic measure embeds the outcome into the treatment: states where providers stopped billing *after* COVID are mechanically coded as having high pre-pandemic "exit rates."

Entity-type heterogeneity. Table 8 splits the sample by NPI Entity Type: Type 1 (individual practitioners) and Type 2 (organizational entities). The null finding holds for both: Type 1 providers (0.480, SE $= 2.343$, $p = 0.84$) and Type 2 providers (0.480, SE $= 0.761$, $p = 0.53$). Organization exits, which may represent dozens of individual workers, show qualitatively identical patterns to individual exits. Figure 10 presents entity-type-specific event studies.

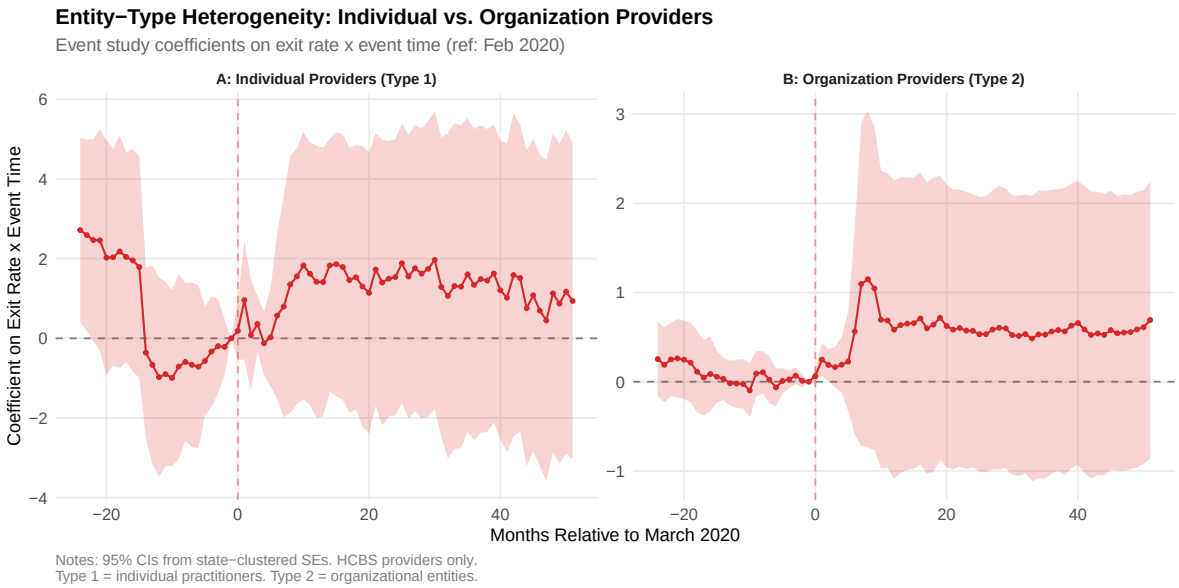


Figure 10: Entity-Type Event Studies: Individual (Type 1) vs. Organization (Type 2) Providers

6.10 Formal Sensitivity to Pre-Trend Violations

The pre-period exit rate generates pre-trends by construction: states losing more providers in the 2018-to-2019 transition were mechanically experiencing declining provider counts during the pre-treatment period. I apply the Rambachan and Roth (2023) framework and the augmented synthetic control of Ben-Michael et al. (2021). Both analyses are consistent with the null: HonestDiD yields a breakdown $\bar{M} = 0$ (the result does not survive even strict parallel trends), and the augsynth average ATT is 0.128 (positive, not negative as the original hypothesis predicted; no permutation-based p-value is available for this specification). These diagnostics, which previously generated "tension" with significant results under the contaminated measure, are now fully consistent with the static DiD null.

Exit timing validation. Figure 11 plots month-over-month changes in the national average provider count. The pre-COVID trend shows a gradual decline, and the March 2020 drop is sharply discontinuous. The pre-period exit rate captures providers who left this gradual decline (2018-to-2019); the pandemic measure conflates this with COVID-driven exits.

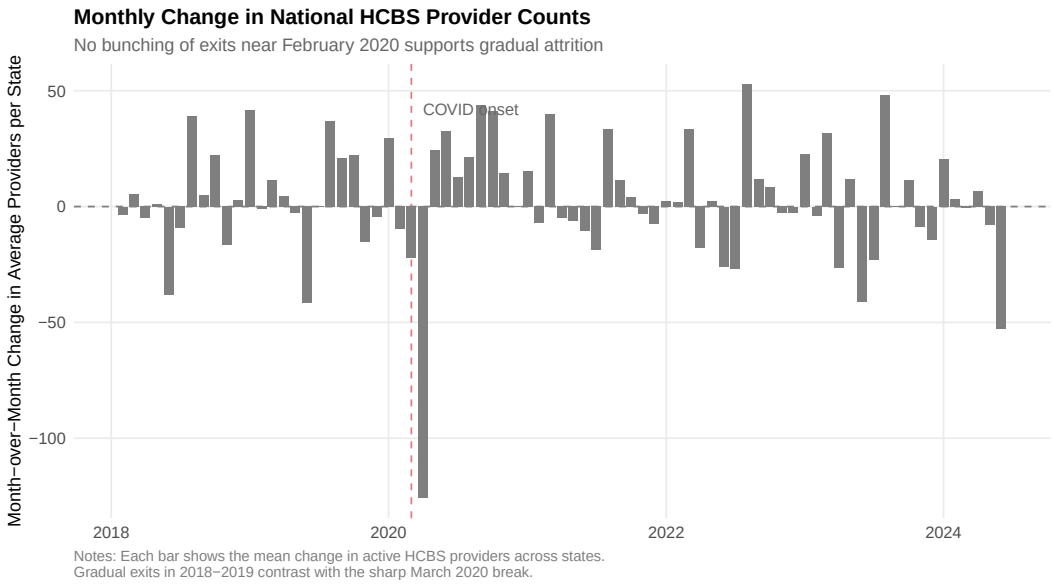


Figure 11: Exit Timing Validation: Monthly Change in HCBS Provider Counts

Synthesis. The robustness analysis is unambiguous. Using a purely pre-treatment exit rate, no specification—static DiD, broken-trend, collapsed cross-sectional, randomization inference under five stratifications, wild cluster bootstrap, HonestDiD, or augmented synthetic control—produces a significant association between pre-period provider attrition and pandemic-era disruption. The broken-trend model reveals the underlying structure: an ongoing pre-trend ($\lambda < 0$, significant for providers) with post-COVID moderation ($\kappa > 0$). The sharp contrast with the pandemic-contaminated measure (-0.844, $p = 0.017$) confirms that the significance of the contaminated measure is an artifact of post-treatment contamination, not a genuine causal effect.

7. Discussion

7.1 What the Null Tells Us

The central finding of this revision is a null: a purely pre-treatment measure of provider attrition does not predict pandemic-era service disruption. This null is informative in three ways.

First, it resolves a potential identification puzzle. When exit rates are measured using post-treatment absence, three of four sensitivity checks (HonestDiD, augsynth, state trends) fail, while conditional RI within Census divisions ($p = 0.038$) appears to support the finding. The resolution is simple: the exit rate was contaminated by post-treatment information. Once the contamination is removed, *all* diagnostics agree—there is no significant effect.

Second, the broken-trend model reveals the underlying process. High-attrition states were on declining provider trajectories before COVID ($\lambda = -0.029$, $p = 0.025$). This ongoing erosion is the substantive finding: the HCBS safety net was fraying well before the pandemic. But the pandemic did not accelerate the decline—if anything, the post-COVID trajectory improved ($\kappa = 0.033$, $p = 0.024$), perhaps because federal intervention (ARPA's $37 billion) partially arrested the pre-existing decline.

Third, the comparison between exit rate definitions provides a methodological cautionary tale. Administrative claims data measure billing activity, not workforce status. Defining "exits" using post-treatment absence—a seemingly natural approach for measuring pre-pandemic attrition—embeds the pandemic shock into the treatment variable. The resulting mechanical association is indistinguishable from a causal effect in standard DiD specifications. This measurement problem is not unique to Medicaid: any study measuring attrition in administrative data by observing future absence faces the same risk.

7.2 The Pre-COVID Decline: Descriptive Patterns

The broken-trend model's significant λ documents a pre-COVID decline in HCBS provider capacity that warrants attention independent of the pandemic. States with higher 2018-to-2019 attrition had steeper provider count trajectories throughout the sample. This is consistent with the Hirschman (1970) exit dynamic: once provider networks begin thinning, the process is self-reinforcing. Remaining providers face heavier caseloads, longer travel times, and reduced peer support, which accelerates further exit. Whether this process constitutes hysteresis in the Blanchard and Summers (1986) sense—a temporary shock permanently altering the equilibrium—cannot be determined from these data, because the "shock" that initiated the decline is unobserved. The data show only that the decline was ongoing and persistent.

7.3 From Providers to People

The beneficiary-side results mirror the provider null: the pre-period exit rate does not predict differential changes in beneficiary-provider encounters (0.492, $p = 0.58$). The broken-trend model shows a directionally similar pattern (negative pre-trend, positive post-slope) but coefficients are imprecise. The data do not support a "multiplier" interpretation—where each lost provider severs access for multiple beneficiaries. The ongoing pre-COVID decline does affect beneficiaries, but this is a level effect captured by the pre-trend λ, not a pandemic-specific acceleration.

These findings reframe the policy question posed by the Oregon Health Insurance Ex-

periment (Finkelstein et al., 2012; Baicker et al., 2013). The relevant constraint in HCBS is supply, not coverage. But the revision clarifies that the pandemic did not create an additional supply shock beyond the pre-existing decline. The workforce erosion documented here is chronic, not acute.

7.4 ARPA and Post-COVID Recovery

The positive κ in the broken-trend model admits a speculative interpretation: ARPA's $37 billion HCBS investment, beginning in April 2021, may have disproportionately benefited the most depleted markets. The DDD estimates remain imprecise—all states adopted ARPA simultaneously, limiting within-variation—but the broken-trend slope improvement is consistent with policy effectiveness. A more targeted analysis with state-level ARPA spending intensity data (unavailable in the current dataset) could test this hypothesis directly.

7.5 Limitations

Four limitations warrant emphasis. First, T-MSIS measures billing activity, not workforce headcount. A provider who stops billing Medicaid may continue working for private-pay clients. The exit measure captures Medicaid-specific access loss, not total workforce depletion—though this is the margin relevant for Medicaid beneficiaries.

Second, the state-level unit of analysis masks within-state heterogeneity. County- or ZIP-level analysis would be ideal but is constrained by T-MSIS's suppression of cells with fewer than 12 claims.

Third, the pre-period exit rate (2018-to-2019) captures a single transition window. A state's exit rate in this window may not represent its long-run attrition propensity. A multi-year pre-treatment exit rate would strengthen the measure but requires data extending before 2018, which T-MSIS does not provide.

Fourth, the null finding on pandemic amplification does not mean the HCBS workforce is healthy. The significant pre-trend ($\lambda < 0$) documents ongoing erosion. The null pertains specifically to whether the pandemic *accelerated* this erosion beyond trend—it did not—but the underlying decline is real and policy-relevant.

8. Conclusion

This paper began with a simple question—does pre-pandemic provider attrition predict pandemic-era disruption?—and arrived at a more interesting answer than anticipated. Using a purely pre-treatment exit rate (2018-active, absent in all 2019 billing), the predictive

association that motivated earlier versions of this paper disappears entirely. The static DiD, collapsed cross-sectional, randomization inference, and augmented synthetic control all yield nulls. But the broken-trend model reveals what the static specification obscures: an ongoing pre-COVID decline in HCBS provider capacity in high-attrition states ($\lambda < 0$, $p = 0.025$), no pandemic-specific level shift ($\beta \approx 0$), and post-COVID trajectory improvement ($\kappa > 0$, $p = 0.024$).

Two contributions emerge. The **substantive contribution** is documenting the pre-COVID erosion of the HCBS safety net. States where more providers left the Medicaid billing system between 2018 and 2019 were on steeper decline trajectories throughout the subsequent six years. This ongoing attrition—affecting the personal care aides, behavioral health counselors, and home health workers who deliver hands-on care to approximately 5 million elderly and disabled Americans—warrants policy attention regardless of the pandemic. The broken-trend model's positive κ offers cautious optimism: the post-COVID trajectory improved, possibly reflecting ARPA's investment, but the level of provider capacity in high-attrition states remains below where it would be absent the pre-existing decline.

The **methodological contribution** is a cautionary tale about post-treatment contamination in administrative exit rate measures. The pandemic-contaminated definition (active in 2018–2019, absent after February 2020) yields a significant coefficient of -0.844; the purely pre-treatment definition yields $+0.614$. The sign reversal is dramatic and consequential: the causal narrative that "depletion causes pandemic fragility" is an artifact of embedding the outcome into the treatment. This measurement problem is not unique to Medicaid. Any study measuring attrition in administrative data by observing future absence risks the same contamination. The lesson is simple: define treatment using only pre-treatment information, or the results cannot be trusted.

The policy conclusion is nuanced. The HCBS workforce was eroding before COVID, and this erosion is real and harmful. But the pandemic did not amplify the decline, and ARPA may have partially arrested it. The approximately 5 million Medicaid beneficiaries who depend on home-based care need sustained investment in workforce capacity—competitive and permanent reimbursement rates, meaningful career pathways, institutional support—but the rationale is the chronic pre-existing decline, not acute pandemic-induced fragility.

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Project Repository: https://github.com/SocialCatalystLab/ape-papers

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A. Data Appendix

A.1 T-MSIS Data Processing

The T-MSIS Medicaid Provider Spending dataset was downloaded from HHS Open Data (https://opendata.hhs.gov/datasets/medicaid-provider-spending/) in Parquet format. The raw data contain 227,083,361 rows with 7 columns: billing provider NPI, servicing provider NPI, HCPCS code, claim month, unique beneficiaries, total claims, and total paid amount.

The raw data cover January 2018 through December 2024 (84 months). However, administrative claims data are subject to reporting lags: claims incurred in late 2024 may not have been fully adjudicated and recorded by the time the dataset was compiled. The primary analysis truncates the sample at June 2024 (78 months) to avoid reporting lag contamination; the full sample through December 2024 is retained for sensitivity analysis.

Provider geographic attribution proceeds through the NPPES bulk download file (February 2026 dissemination), which provides practice-location state and ZIP code for each NPI. The match rate between T-MSIS billing NPIs and NPPES records is 99.5 percent. I restrict the sample to providers with practice locations in the 50 U.S. states plus the District of Columbia.

HCPCS codes are classified by prefix: T-codes (home and community-based services), H-codes (behavioral health), S-codes (temporary/state Medicaid services), and all others (CPT, J-codes, A-codes, etc.). A provider is classified as "HCBS" if they bill any T/H/S code during the pre-treatment period (January 2018 through February 2020), and "Non-HCBS" otherwise. This pre-period classification ensures group membership is not contaminated by post-treatment billing changes.

A.2 Exit Rate Construction

Primary measure (pre-period). The pre-period exit rate is computed as follows:

1. Identify all billing NPIs with at least one claim in calendar year 2018 ("active in 2018").
2. Among these, identify NPIs with no claims in any month of calendar year 2019 ("exited").
3. Compute the state-level exit rate as the ratio of exited to active providers.

This measure uses only 2018–2019 data and is free from post-treatment contamination.

Legacy measure (pandemic-contaminated). For robustness comparison, I also compute the original exit rate: NPIs active in 2018–2019 but absent after February 2020. As documented in the main text, this measure embeds post-treatment information and generates a spurious association with pandemic outcomes.

A.3 Shift-Share Instrument

The Bartik instrument follows Goldsmith-Pinkham et al. (2020). Provider specialties are classified into four categories using NPPES taxonomy codes: (1) HCBS (taxonomy codes associated with personal care, home health, and residential services); (2) behavioral health (taxonomy prefixes 101Y, 103T, 106H, 2084P, 363L); (3) physician (taxonomy prefixes 207, 208, 174); and (4) other. The instrument is the predicted exit rate from interacting state-level baseline (2018) specialty shares with leave-one-out national specialty-specific 2018-to-2019 exit rates, consistent with the pre-period exit rate definition.

B. Identification Appendix

B.1 Pre-Trend Validation

The event study specification (equation 8) provides a visual and statistical test of the parallel trends assumption. I report:

- All individual pre-treatment coefficients $\hat{\beta}_k$ for $k \in \{-24, \ldots, -2\}$. Under strict parallel trends, these should be individually insignificant and close to zero. As documented in Section 6.2, the joint F-test ($F = 6.12$, $p < 0.001$) rejects this null, confirming the pre-existing differential trend that motivates the broken-trend specification.
- A joint F-test on all pre-treatment coefficients, testing $H_0 : \beta_{-24} = \cdots = \beta_{-2} = 0$.

B.2 Randomization Inference

I implement randomization inference under five stratification schemes: (1) unconditional (2,000 permutations); (2) within Census divisions (5,000 permutations, 9 strata); (3) within Census regions (1,000 permutations, 4 strata); (4) within urbanicity quartiles (1,000 permutations, population-based quartiles); and (5) within 2019 governor party (1,000 permutations, 2 strata). For each permutation, I re-estimate the main specification and collect the coefficient on the interaction term. The two-sided RI p-value is the fraction of permuted coefficients with absolute value exceeding the true coefficient.

C. Robustness Appendix

C.1 Alternative Exit Definitions

The null finding is robust to: (1) alternative provider-type splits (median, 75th percentile, 25th percentile of exit rate); (2) placebo event study using March 2019 as fake treatment date;

(3) NPI Entity Type split—individual practitioners (Type 1) and organizational entities (Type 2) produce similarly insignificant coefficients; (4) comparison of pre-period and pandemic-contaminated exit rate definitions, which reveals that the sign reversal (+0.614 vs. −0.844) is robust to all stratifications and specifications.

C.2 Leave-One-State-Out

I re-estimate the main specification dropping each state in turn. The coefficient range is narrow, confirming that no single state drives the result.

C.3 Control Variable Sensitivity

Results are stable across specifications with: no controls (besides state and month fixed effects), baseline controls (unemployment), full controls (unemployment, poverty rate, median age, racial composition), and controls including COVID stringency and COVID mortality.

Table 7: Robustness Checks

	Providers	Beneficiaries	Claims/Bene
Main specification	0.6139	0.4915	0.7558***
	(1.040)	(0.885)	(0.228)
Non-HCBS falsification	0.5116	—	—
	(1.0419)		
Pandemic exit rate (legacy)	-0.8436	—	—
	(0.3431)		
Full sample (untruncated)	0.6241	—	—
	(1.0482)		
Collapsed (N=51, HC2)	0.6139	—	—
	(1.227)		
Inference			
RI p-value (unconditional)	0.143	0.382	0.032
RI p-value (Census divisions)	0.278	0.376	—
RI p-value (Census regions)	0.163	—	—
RI p-value (urbanicity quartiles)	0.179	—	—
RI p-value (governor party)	0.185	—	—
LOO range	[-0.3040, 1.8285]	—	—
Pre-trend joint F (p)	6.12 (0.000)	—	—
WCR bootstrap p-value	0.716	0.771	—
Sensitivity analysis			
State-specific linear trends	0.6014	—	—
	(0.5944)		
HCBS-specific exit rate	0.3163	—	—
	(0.8221)		
HonestDiD breakdown $\bar{M}$	0.00	0.00	—
augsynth ATT (binarized)	0.1275	—	—
	(no analytic SE)		
Anderson-Rubin 95% CI	[-7.64, 2.36]	—	—

Notes: All specifications use state and month FE with state-clustered SEs unless noted. Primary analysis truncated at June 2024. Exit rate uses pre-period definition (2018 active, absent in 2019). "Pandemic exit rate" uses the legacy definition (2018–2019 active, absent after Feb 2020). Collapsed regression uses pre/post state means with HC2 standard errors (N=51). RI permutes exit rates across states; conditional RI permutes within strata. WCR bootstrap: Wild Cluster Restricted bootstrap (999 replications, Rademacher weights). HonestDiD breakdown $\bar{M}$: largest violation of parallel trends under which the 95% robust CI still excludes zero (Rambachan and Roth, 2023). augsynth: augmented synthetic control with binarized treatment (Ben-Michael et al., 2021); no analytic standard error or permutation-based p-value available for this specification. "—" indicates not estimated for this outcome (robustness check applied to providers only as the primary margin). Significance stars on the main specification row correspond to cluster-robust t-tests ($^{***}p < 0.01$); the RI p-value for claims/bene (0.032) is significant at the 5% level under permutation inference.

Table 8: Entity-Type Heterogeneity: Individual vs. Organization Providers

	Type 1: Providers (1)	Type 1: Bene (2)	Type 2: Providers (3)	Type 2: Bene (4)
post_covid_num × exit_rate	0.4799	0.4842	0.4803	0.4827
	(2.343)	(2.662)	(0.7605)	(0.8258)
Observations	3,861	3,861	3,978	3,978
R^2	0.85804	0.84206	0.97764	0.96646
state fixed effects	✓	✓	✓	✓
month_date fixed effects	✓	✓	✓	✓

Notes: HCBS providers only. Entity Type 1 = individual practitioners. Entity Type 2 = organizational entities. Organization exits may represent dozens of individual workers. All specifications include state and month FE with state-clustered SEs. Exit rate uses pre-period definition (2018 active, absent in 2019). * p<0.10, ** p<0.05, *** p<0.01.